

Chapter I

Problems

1 INPhO-2006

Indian National Physics Olympiad - 2006

INPhO-2006

Jan. 29, 2006

Maximum Marks: 90

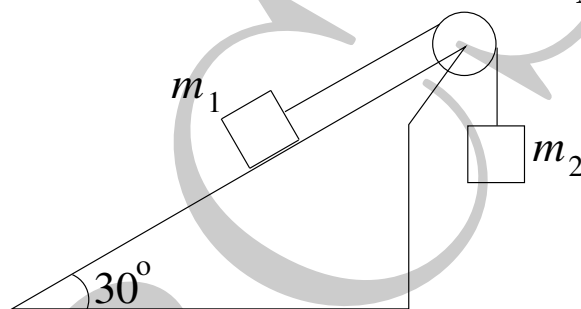


Figure 1: Problem 1

1. In the diagram shown (Fig. (1)), $m_1 = 1$ kg, $m_2 = 1$ kg and coefficient of friction, both static and dynamic, between m_1 and plane is $\mu = 0.6$. The two masses are connected by a light inextensible string passing over a light frictionless pulley. Take $g = 10 \text{ m}\cdot\text{s}^{-2}$.

- (a) Find the acceleration of the system.
- (b) Find the force of friction and the magnitude of the tension in the string.

[10]

2. A block of uniform mass M is at rest on a table. A disk of mass $2M$, radius R and of the same height as the block, which is initially spinning about its axis with angular speed ω_0 , is placed on the table such that it touches the block (see Fig. (2)). The block – disk system starts moving such that they are in contact throughout the motion. Coefficient of friction, both kinetic and static, between the table and block and between the table and disk is μ . Friction between disk and the block may be ignored.

- (a) Obtain an expression for the initial acceleration of the block – disk system.

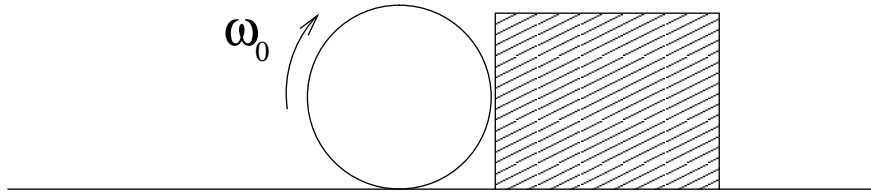


Figure 2: Problem 2

- (b) Obtain an expression for the time t^* at which pure rolling (i.e. rolling without slipping) starts.
- (c) Obtain an expression for the total time t_{tot} in which the block comes to the rest. Assume that pure rolling persists for $t > t^*$.

[14]

3. An ideal gas goes through a reversible cycle which consists of two isobaric and two adiabatic processes as shown in the $P - V$ diagram (Fig. (3)).

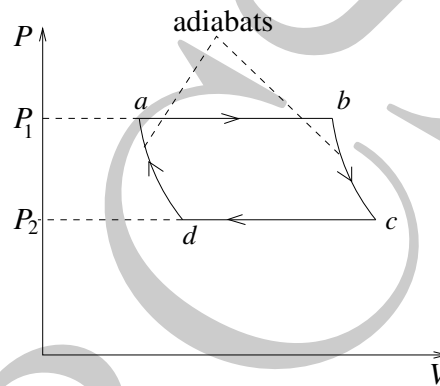


Figure 3: Problem 3

- (a) Obtain an expression for the efficiency of the cycle in terms of the temperatures $\{T_a, T_b, T_c, T_d\}$.
- (b) Obtain an expression for the efficiency of the cycle in terms of the pressures $\{P_1, P_2\}$ and γ . Here γ is the ratio of the specific heat at constant pressure and specific heat at constant volume.
- (c) Draw the equivalent $V - T$ diagram for this cycle.
[Note: V along y -axis and T along x -axis.]
- (d) State the expression for the corresponding Carnot cycle working with the same gas and between the highest and lowest temperatures defined by the above cycle. Which of these two cycles has the higher efficiency?

[12]

4. A thin plano-convex lens of radius $R = 10$ cm, refractive index $\mu_2 = 1.5$ has its curved surface in liquid of refractive index $\mu_3 = 1.2$ and the plane surface exposed to air of refractive index $\mu_1 = 1.0$. A self luminous particle oscillating simple harmonically with small amplitude $\sqrt{2}$ cm is placed on the axis of the lens as shown in Fig. (4). Determine the orientation, amplitude and phase

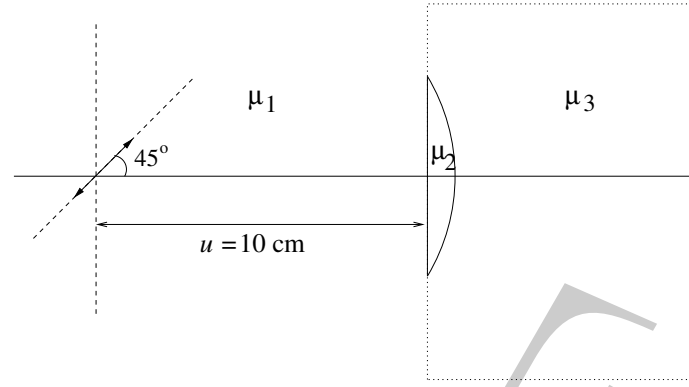


Figure 4: Problem 4

difference of the oscillating final image with respect to the object.

[8]

5. A thin circular disk of radius R is uniformly charged with charge σ ($\sigma > 0$) per unit area. The disk rotates about its axis OX with a uniform angular speed ω (see Fig. (5)). A small magnetic dipole of moment $\vec{\mu}$ is located at $P(a, 0, 0)$ on the axis of the disk ($a > 0$).

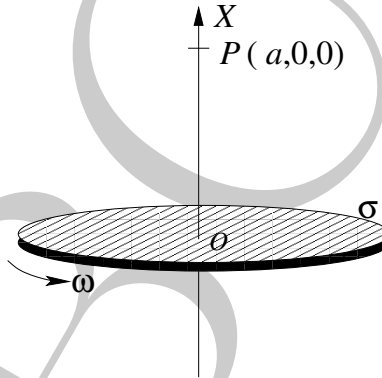


Figure 5: Problem 5

- Obtain the expression for the magnetic moment of the disk?
- Obtain the expression for the magnetic field $\vec{B}$ due to the rotating disk at P ?
- Obtain the approximate expression for $\vec{B}$ when $a \gg R$.
- Obtain the force on the dipole placed at P given that $a \gg R$.

[Note: You can use the formula for $\vec{B}$ on the axis of a circular current, namely

$$|\vec{B}| = \frac{\mu_0}{4\pi} \frac{2i\pi r^2}{(r^2 + x^2)^{3/2}} \quad]$$

[14]

6. A 1.00 kW cylindrical (monochromatic) laser light beam of radius δ is used to levitate a solid aluminium sphere of radius R by focusing it on the sphere

from below (see Fig. (6)). The laser light is reflected by the aluminium sphere without any absorption.

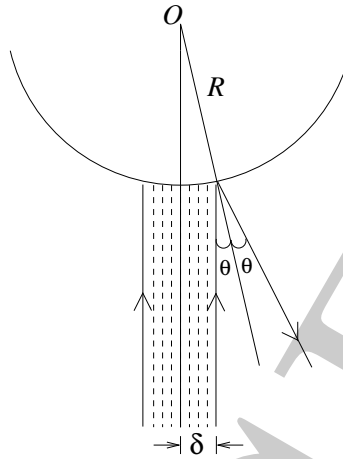


Figure 6: Problem 6

- (a) Take the momentum of each photon in the light beam to be p . Express the force exerted on the aluminium sphere by the beam in terms of p , δ , R , and n where n is the number of photons per unit area per unit time.
- (b) Now consider the special case $\delta \ll R$. Calculate the mass of the sphere, assuming that it floats freely on the light beam?

[Hint: Part (b) can be done independently of Part (a)]

[10]

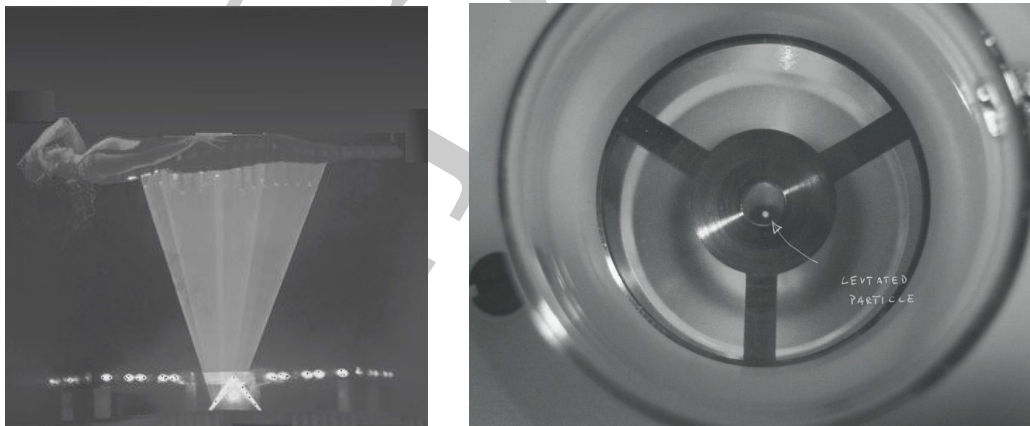


Figure 7: Demonstrations of laser levitation

7. An electron in the Li^{++} ion makes a transition from $n = 4$ to $n = 3$ state.
 - (a) Find the wavelength of emitted photon in this process. To what region of the electro-magnetic spectrum does this wavelength belong?

- (b) This photon impinges on a photoelectric sensitive metal having work function 3.20 eV. Calculate the maximum kinetic energy and the corresponding de Broglie wavelength of emitted photo-electron.

(Ionization energy of hydrogen atom = 13.60 eV) [8]

8. Lyttleton - Bondi Model for the Expansion of the Universe*

In 1959 Lyttleton and Bondi suggested that the expansion of the Universe could be explained on the basis of Newtonian mechanics if matter carried a net electric charge. Imagine a spherical volume of astronomical size and radius R containing un-ionized atomic hydrogen gas of uniform density $\eta (= 10^{-26} \text{ kg} \cdot \text{m}^{-3})$, and assume that the proton charge $e_p = -(1 + y)e$, where e is the electron charge.

- Obtain the value of y for which the electrostatic repulsion becomes larger than the gravitational attraction and the gas expands.
- Obtain an expression for the force of repulsion on an atom which is at a distance R from the centre of the spherical volume. Hence show that the radial velocity is proportional to R . Let us label the proportionality constant as H . Assume that the density is maintained constant by the continuous creation of matter in space. Assume also that the value of y is larger than the equilibrium value calculated in part (a) above and hence ignore gravity.
- Calculate the numerical value of H . Take the value of y to be one order of magnitude larger than the equilibrium value calculated in part (a) above.
- Given that at time $t = 0$, the volume of the Universe was V_0 , obtain an expression for the volume expansion of the Universe.
- Why do you think the Lyttleton - Bondi model has been largely discarded by the scientific community?

[14]

*Ref. R.A.Lyttleton and H.Bondi, Proceedings of Royal Society of London, Volume **A 252**, page 313 - 333, (1959)



Raymond Arthur Lyttleton (7 May 1911-16 May 1995) : English mathematician and theoretical astronomer who researched stellar evolution and composition. In 1939, with Fred Hoyle, he demonstrated the large scale existence of interstellar hydrogen, refuting the existing belief that space was devoid of interstellar gas. Together, in the early 1940's, they applied nuclear physics to explain how energy is generated by stars. In his own monograph (1953) Lyttleton described stability of rotating liquid masses, which he extended later to explain that

the Earth had a liquid core resulting from a phase change associated with a combination of intense pressure and temperature. With Hermann Bondi, in 1959, he proposed the electrostatic theory of the expanding universe. He authored various astronomy books. One of them "Mysteries of the Solar System", was co-authored with Edwin Land and was quite popular.